Ashvin Gupta

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Second-year PhD Researcher at Imperial College London focused on developing Transformer models for high-dimensional data modelling and cancer diagnosis and post-training using reinforcement learning. Expertise in AI for Healthcare with prior industrial experience in Computer Vision. Proven ability in teaching mathematical and computational modules with interest in broadening horizons.

Education

Imperial College London	Ph.D. in AI and Machine Learning	2024 – 2027
Imperial College London	Biomedical Engineering, First Class Honours, Dean's List Year 1-3	2019 – 2023
Hymers College	A-Levels, 4 A*s	2009 – 2019

Professional Experience

Oxford Heartbeat 2023 – 2024
Machine Learning Engineer

- Developed a **supervised deep-learning** thresholding algorithm by **fine-tuning a ResNet18** architecture to segment vessels within DICOM scans of the brain using **PyTorch**, **Pandas** and **Scikit-Learn**.
- Used **nnU-Net** to segment areas within a 3D scan that show high probabilities of exhibiting 'bleeding' that now assists with centreline extraction.
- Created a **GUI** designed to be used by clinicians to label vessels within CT scans of the brain using image processing techniques that reduced the labelling times from many hours to minutes. Presented the project to Innovate UK who granted extra funding.
- Completed **co-registration** of MRI scans and CT scans for MRI integration into the software.

Bank of America Summer 2022
Software Engineer Summer Intern

- Set up a **CI/CD** pipeline that would build and run the most recent version of the team's application, completing stress testing nightly. Job would identify irregularities and alert **QA** immediately, mitigating risk. Worked in an **Agile** methodology, using **Git** commands to collaborate on large-scale code. Declined return offer.

Project Experience

Diagnosing Cancers from high dimensional data 2025 – Present
Supervisors: Alessandra Russo, Brendan Delaney

- Developing **transformer-based encoder** and **decoder** models for diagnosing risk of cancer, given their electronic health record as context. Investigating both **classification** and **generation** tasks through **fine-tuning LLMs** using **LoRA** and **GRPO** for post-training explanations using the college's **HPC** for job submissions.

Converting NICE guidelines to a Computer Readable Format 2025
Supervisors: Alessandra Russo, Brendan Delaney

- Built a pipeline that uses **LLMs** with **in-context learning** to translate medical guidelines to an **Answer Set Model**. Model can be appended with patient features and solved for automatic, patient-specific recommendations.

Physics-Informed Neural Networks (PINNs) for Atrial Fibrillation 2023
Supervisors: Marta Varela

- Used **PINNs** to solve PDEs in that govern action potential propagation. Evaluated that PINNs can accurately replicate electrical properties in 3 dimensions for varying action potentials using less than **10% of data** for training.

Teaching

Mathematics for Machine Learning 2025

- Teaching Assistant for Master's level course that covers vector calculus, statistics and probability theory.

Computing Research Collective 2025

- Leading sessions with five BEng Computing students on how to read and criticise research papers, exploring areas relating to AI in Healthcare. Giving focused feedback on student presentations and mentorship for further study.

Publications

1. **Gupta A,** Prociuk D, Delaney B, Russo A "Automatic Conversion of NICE Guidelines to an Executable Computational Model using Large Language Models" (Learning Health Systems 2026) *Expected*
2. Varela M, **Gupta A**, Martin C, Oved A, Chiu A, Peters S N, Bharath A A "Physics-Informed Neural Networks can accurately model cardiac electrophysiology in 3D geometries and fibrillatory conditions" (STACOM 2024)